

Extending Swanepoel to $m = 7$ and pushing the bound to $g(n) \geq \lceil 7n/27 \rceil$

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Abstract

A Euclidean minimum-distance graph is the unit-distance graph of a finite point set in $\mathbb{R}^2$ whose pairwise distances are all at least 1. Let $g(n)$ be the minimum, over all such graphs on n vertices, of the independence number. This document gives a complete proof, relative to Swanepoel's published broken-lattice inputs, that every Euclidean minimum-distance graph G satisfies

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Equivalently,

$$g(n) \geq \left\lceil \frac{7n}{27} \right\rceil.$$

The proof specializes Swanepoel's broken-lattice input to $m = 7$, isolates the unique equality break, proves the required boundary-side orientation, and eliminates every remaining local configuration by explicit independent-set certificates.

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1. Definitions and theorem

Definition 1 (Euclidean minimum-distance graph). *A Euclidean minimum-distance graph is a graph G obtained from a finite point set $P \subset \mathbb{R}^2$ satisfying*

$$|x - y| \geq 1 \quad (x \neq y, x, y \in P),$$

with vertex set P and an edge xy exactly when

$$|x - y| = 1.$$

For $S \subseteq V(G)$, write

$$N(S) = \{v \in V(G) \setminus S : v \text{ is adjacent to some } s \in S\}.$$

Definition 2. *Let $g(n)$ be the minimum of $\alpha(G)$ over all Euclidean minimum-distance graphs G on n vertices.*

Theorem 1. *Relative to the Swanepoel inputs stated in Proposition 1, every Euclidean minimum-distance graph G satisfies*

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Consequently,

$$\alpha(G) \geq \left\lceil \frac{7|V(G)|}{27} \right\rceil$$

and

$$g(n) \geq \left\lceil \frac{7n}{27} \right\rceil.$$

2. Published Swanepoel input

The proof imports exactly the following Swanepoel facts from [1]. The notation follows Swanepoel's broken-lattice proof.

Proposition 1 (Swanepoel broken-lattice input). *The boundary-arc structure is Theorem 4 on p. 661; the two-extra-neighbour statement is Lemma 8 on pp. 659–660; the triple obstruction and the cap used below come from Lemma 9 and its proof on pp. 660–661; and the Euclidean all-but-one equality and break inequality are Lemma 10 and inequality (5) on pp. 661–665.*

Let $m \geq 5$, and let G be a smallest counterexample to

$$\alpha(G) \geq \frac{m}{4m-1}|V(G)|.$$

Then there is a nonconcave boundary arc

$$p_0, p_1, \dots, p_{2m-3}$$

with total turn less than $\pi/3$, with

$$\deg(p_0) = 3, \quad \deg(p_i) = 4 \quad (1 \leq i \leq 2m-3),$$

and with a unique common neighbour q_i of p_i and p_{i+1} for each boundary edge $p_i p_{i+1}$.

For each

$$1 \leq i \leq 2m - 5,$$

Swanepoel's Lemma 8, pp. 659–660, gives exactly two extra neighbours r_i, s_i of q_i outside the local set

$$\{p_i, p_{i+1}, q_{i-1}, q_{i+1}\},$$

where unavailable endpoint q -terms are omitted.

If

$$s_i = r_{i+1} \quad (i = a, a + 1, a + 2),$$

then Swanepoel's Lemma 9, pp. 660–661, gives

$$a \geq 2m - 10.$$

Moreover, the proof of Lemma 9 on pp. 660–661 gives the cap

$$|N(s_{a+1}) \setminus \{s_a, s_{a+2}, q_{a+1}, q_{a+2}\}| \leq 2.$$

In the Euclidean plane, Swanepoel's Lemma 10 on pp. 661–665 gives

$$s_i = r_{i+1} \quad (1 \leq i \leq 2m - 6)$$

for all but at most one index i . If

$$s_i \neq r_{i+1},$$

then Lemma 10, specifically inequality (5), gives the break inequality

$$\tau_i + 2\tau_{i+1} + \tau_{i+2} \geq \frac{\pi}{3}.$$

Remark 1. The proof below proves the theorem relative to Proposition 1. The endpoint use of q_1 and q_9 uses Lemma 8 in the range $1 \leq i \leq 2m - 5$. The cap for t_5, t_6, t_7 is the cap obtained in the proof of Lemma 9, not a separate theorem statement. The uniform side formula for q_i is not imported from Swanepoel; it is proved internally in Lemma 5.

3. Minimal-counterexample deletion criterion

Lemma 1 (Deletion criterion). Let G be a smallest counterexample to

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

If $\emptyset \neq S \subseteq V(G)$ is independent and

$$|N(S)| \leq \left\lfloor \frac{20|S|}{7} \right\rfloor,$$

then G is not a counterexample.

Proof. Let

$$n = |V(G)|.$$

Set

$$G' = G - (S \cup N(S)).$$

Since $S \neq \emptyset$, the graph G' has fewer vertices than G . By minimality of G ,

$$\alpha(G') \geq \frac{7}{27}|V(G')|.$$

Since no vertex of G' is adjacent to S , an independent set in G' can be joined with S . Therefore

$$\alpha(G) \geq |S| + \alpha(G').$$

Also

$$|V(G')| = n - |S| - |N(S)|.$$

Hence

$$\begin{aligned} \alpha(G) &\geq |S| + \frac{7}{27}(n - |S| - |N(S)|) \\ &= \frac{7n}{27} + \frac{20|S| - 7|N(S)|}{27}. \end{aligned}$$

If $|N(S)| \leq 20|S|/7$, then the last numerator is nonnegative. Thus

$$\alpha(G) \geq \frac{7n}{27},$$

contradicting that G is a counterexample. □

4. The $m = 7$ broken-lattice setup

Assume, for contradiction, that G is a smallest counterexample to

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Apply Proposition 1 with $m = 7$. Then

$$2m - 3 = 11, \quad 2m - 6 = 8, \quad 2m - 10 = 4.$$

Thus the boundary arc is

$$p_0, p_1, \dots, p_{11}.$$

Lemma 2 (The unique $m = 7$ break). *Exactly one equality among*

$$s_i = r_{i+1} \quad (1 \leq i \leq 8)$$

fails, namely

$$s_3 \neq r_4.$$

Consequently,

$$s_i = r_{i+1} \quad i = 1, 2, 4, 5, 6, 7, 8.$$

Proof. By Proposition 1, at most one equality among

$$s_i = r_{i+1} \quad (1 \leq i \leq 8)$$

fails. If three consecutive equalities begin at a , then $a \geq 4$. Therefore true triples beginning at $a = 1, 2, 3$ are impossible. These forbidden triples are

$$(1, 2, 3), \quad (2, 3, 4), \quad (3, 4, 5).$$

Since the triple $(1, 2, 3)$ cannot be true, at least one equality fails. Since at most one equality fails, exactly one equality fails. To make all three forbidden triples false, the single failure must meet all three triples. Their only common index is 3. Hence

$$s_3 \neq r_4.$$

□

Whenever $s_i = r_{i+1}$, write

$$t_i = s_i = r_{i+1}.$$

Thus t_i is defined for

$$i = 1, 2, 4, 5, 6, 7, 8.$$

5. Boundary-side orientation

Lemma 3 (Straight-line planarity). *The straight-line drawing of a Euclidean minimum-distance graph has no crossing pair of edges.*

Proof. Suppose two unit edges ab and cd cross at an interior point x . One endpoint of ab , say u , satisfies

$$|u - x| \leq \frac{1}{2},$$

and one endpoint of cd , say v , satisfies

$$|v - x| \leq \frac{1}{2}.$$

The edges are distinct and cross properly, so $u \neq v$. If equality held in

$$|u - v| \leq |u - x| + |x - v| \leq 1,$$

then u, x, v would be collinear and both distances would be exactly $1/2$, which is incompatible with two distinct straight segments crossing properly. Hence

$$|u - v| < 1,$$

contradicting the minimum-distance hypothesis. □

Lemma 4 (Empty unit triangle). *Let abc be a unit equilateral triangle whose vertices are in the point set. No other vertex of the point set lies in the closed triangular region bounded by abc .*

Proof. The closed triangular region has diameter 1. If a distinct point z lay in it, then the minimum-distance condition would require

$$|z - a| \geq 1, \quad |z - b| \geq 1, \quad |z - c| \geq 1.$$

Since the region has diameter 1, all three distances would have to be exactly 1. The only point in the closed triangle at distance 1 from both a and b is c . Thus no distinct point z exists. □

Lemma 5 (Boundary-side orientation). *Let $p_i p_{i+1}$ be one of the boundary triangle edges in the Swanepoel arc, and let q_i be the unique common neighbour of p_i and p_{i+1} . Orient the boundary cycle so that the interior of the drawing is locally on the left of each directed boundary edge in the arc. Then q_i lies on the interior side of the directed edge $p_i p_{i+1}$. Consequently, after choosing the complex coordinate orientation so that left rotation by 60° is multiplication by $\omega = e^{i\pi/3}$,*

$$q_i = p_i + \omega(p_{i+1} - p_i)$$

for every relevant i in the arc.

Proof. Write $a = p_i$, $b = p_{i+1}$, and $q = q_i$. Since q is adjacent to both endpoints of the unit edge ab ,

$$|a - b| = |a - q| = |b - q| = 1.$$

Thus a, b, q form a unit equilateral triangle, and q is one of the two equilateral apices on the segment ab .

The segment ab is an outer-boundary edge of the straight-line plane drawing. One side of ab is incident with the outer face, and the other side is the interior side of the drawing. Suppose, for contradiction, that q lies on the outer-face side of ab . Then the triangular cycle $aqba$ lies on the outer-face side of ab . By Lemma 4, its closed triangular region contains no other graph vertex. By Lemma 3, no graph edge crosses its boundary. Therefore the bounded triangular region enclosed by a, q, b is a face of the plane embedding. On that side the edge ab is then incident with this bounded triangular face, not with the outer face. This contradicts that ab is an outer-boundary edge with the outer face on that side.

Hence q lies on the interior side of ab . By the chosen orientation of the boundary, the interior side is the left side of the directed edge ab . The left equilateral apex of the directed unit segment from a to b is

$$a + \omega(b - a).$$

Thus

$$q_i = p_i + \omega(p_{i+1} - p_i).$$

□

6. Coordinate conventions and elementary geometry

Throughout, identify $\mathbb{R}^2$ with $\mathbb{C}$ according to Lemma 5, and put

$$\omega = e^{i\pi/3}.$$

Write

$$e_i = p_{i+1} - p_i, \quad |e_i| = 1.$$

By Lemma 5,

$$q_i = p_i + \omega e_i.$$

Let τ_{i+1} denote the clockwise turn from e_i to e_{i+1} , so

$$e_{i+1} = e_i e^{-i\tau_{i+1}}, \quad \tau_{i+1} \geq 0.$$

The total turn of the arc is less than $\pi/3$. This convention is local: only turn sums involving existing consecutive edges are used below.

Lemma 6 (Degree bound). *Every vertex of a Euclidean minimum-distance graph has degree at most 6.*

Proof. Let v be a vertex. Its neighbours lie on the unit circle centred at v . If x, y are two distinct neighbours and θ is their smaller angular separation around v , then

$$|x - y| = 2 \sin(\theta/2).$$

Since all distinct vertices are at distance at least 1,

$$2 \sin(\theta/2) \geq 1,$$

and hence

$$\theta \geq \pi/3.$$

At most six directions around a circle can be pairwise separated by at least $\pi/3$. Therefore $\deg(v) \leq 6$. $\square$

Lemma 7 (General $q_i q_{i+1}$ turn criterion). *For every relevant i ,*

$$q_i q_{i+1} \in E(G) \iff \tau_{i+1} = 0.$$

This statement does not assume $s_i = r_{i+1}$.

Proof. Using $q_i = p_i + \omega e_i$,

$$\begin{aligned} q_{i+1} - q_i &= p_{i+1} + \omega e_{i+1} - p_i - \omega e_i \\ &= e_i + \omega e_{i+1} - \omega e_i \\ &= (1 - \omega) e_i + \omega e_{i+1} \\ &= \bar{\omega} e_i + \omega e_{i+1}. \end{aligned}$$

The two summands are unit vectors. Since e_{i+1} is obtained from e_i by a clockwise turn τ_{i+1} , the angle between $\bar{\omega} e_i$ and ωe_{i+1} is

$$\frac{2\pi}{3} - \tau_{i+1}.$$

Because $0 \leq \tau_{i+1} < \pi/3$, this angle lies in $(\pi/3, 2\pi/3]$. A sum of two unit vectors has length 1 exactly when the angle between them is $2\pi/3$. Therefore

$$|q_{i+1} - q_i| = 1 \iff \tau_{i+1} = 0.$$

Since edges are exactly unit distances, the result follows. $\square$

Lemma 8 (Equality-strip formula and $t_i t_{i+1}$ criterion). *If $s_i = r_{i+1}$, then*

$$t_i = p_i + \omega(e_i + e_{i+1}).$$

If in addition both t_i and t_{i+1} are defined, then

$$t_i t_{i+1} \in E(G) \iff \tau_{i+1} + \tau_{i+2} = 0.$$

Proof. The two unit-circle intersections of q_i and q_{i+1} are p_{i+1} and

$$q_i + q_{i+1} - p_{i+1}.$$

On an equality strip this second common neighbour is t_i . Thus

$$\begin{aligned} t_i &= q_i + q_{i+1} - p_{i+1} \\ &= (p_i + \omega e_i) + (p_{i+1} + \omega e_{i+1}) - p_{i+1} \\ &= p_i + \omega(e_i + e_{i+1}). \end{aligned}$$

Now suppose t_i, t_{i+1} are defined. Then

$$\begin{aligned} t_{i+1} - t_i &= p_{i+1} + \omega(e_{i+1} + e_{i+2}) - p_i - \omega(e_i + e_{i+1}) \\ &= e_i - \omega e_i + \omega e_{i+2} \\ &= \bar{\omega} e_i + \omega e_{i+2}. \end{aligned}$$

The angle between these two unit summands is

$$\frac{2\pi}{3} - (\tau_{i+1} + \tau_{i+2}).$$

Thus the length is 1 exactly when this angle is $2\pi/3$, equivalently when

$$\tau_{i+1} + \tau_{i+2} = 0.$$

□

Corollary 1. *If $t_4 t_5 \in E(G)$, then*

$$\tau_5 = \tau_6 = 0.$$

After this, the following cases are exhaustive:

$$\tau_7 > 0,$$

$$\tau_7 = 0, \quad \tau_8 + \tau_9 > 0,$$

and

$$\tau_7 = \tau_8 = \tau_9 = 0.$$

Proof. By Lemma 8,

$$t_4 t_5 \in E(G) \iff \tau_5 + \tau_6 = 0.$$

Since both turns are nonnegative, this is equivalent to $\tau_5 = \tau_6 = 0$. The trichotomy is then the exhaustive trichotomy for the later nonnegative turns τ_7, τ_8, τ_9 . □

Lemma 9 (Forward cone estimate). *Rotate so that $e_a = 1$. For every $j \geq a$, there exists $\phi_j \in [0, \pi/3)$ such that*

$$e_j = e^{-i\phi_j}.$$

Consequently,

$$\Re(e_j) > \frac{1}{2}, \quad \Re(\omega e_j) \geq \frac{1}{2}.$$

Proof. All turns are nonnegative and the total turn is less than $\pi/3$. Hence every later direction is obtained from e_a by a clockwise turn $\phi_j \in [0, \pi/3)$. Therefore

$$e_j = e^{-i\phi_j}.$$

Thus

$$\Re(e_j) = \cos \phi_j > \frac{1}{2}.$$

Also

$$\omega e_j = e^{i(\pi/3 - \phi_j)},$$

and $0 \leq \pi/3 - \phi_j \leq \pi/3$, giving

$$\Re(\omega e_j) \geq \frac{1}{2}.$$

□

Lemma 10 (Triangular-lattice unit vectors). *Let*

$$z = a + b\omega, \quad a, b \in \mathbb{Z}.$$

Then

$$|z|^2 = a^2 + ab + b^2.$$

Moreover, $|z| = 1$ if and only if

$$z \in \{\pm 1, \pm \omega, \pm \omega^2\}.$$

Proof. Since $\omega + \bar{\omega} = 1$,

$$\begin{aligned} |a + b\omega|^2 &= (a + b\omega)(a + b\bar{\omega}) \\ &= a^2 + ab(\omega + \bar{\omega}) + b^2 \\ &= a^2 + ab + b^2. \end{aligned}$$

If $a^2 + ab + b^2 = 1$, then

$$a^2 + ab + b^2 = \left(a + \frac{b}{2}\right)^2 + \frac{3}{4}b^2.$$

Thus $|b| \leq 1$. If $b = 0$, then $a = \pm 1$. If $b = 1$, then $a^2 + a = 0$, so $a = 0$ or $a = -1$, giving ω or ω^2 . If $b = -1$, then $a^2 - a = 0$, so $a = 0$ or $a = 1$, giving $-\omega$ or $-\omega^2$. This gives exactly the six listed units. □

7. Boundary and endpoint neighbourhoods

Lemma 11 (Boundary neighbourhoods). *For $1 \leq i \leq 10$,*

$$N(p_i) = \{p_{i-1}, p_{i+1}, q_{i-1}, q_i\}.$$

Also, for a vertex p_{-1} ,

$$N(p_0) = \{p_{-1}, p_1, q_0\},$$

and

$$N(p_{11}) \subseteq \{p_{10}, q_{10}\} \cup U_{11}, \quad |U_{11}| \leq 2.$$

Proof. For $1 \leq i \leq 10$, Swanepoel gives $\deg(p_i) = 4$. The four displayed vertices are neighbours of p_i . We verify that they are distinct.

First,

$$p_{i+1} - p_{i-1} = e_{i-1} + e_i.$$

If $p_{i+1} = p_{i-1}$, then $e_i = -e_{i-1}$, requiring a turn of π , impossible since total turn is $< \pi/3$.

Next,

$$q_{i-1} = p_{i-1} + \omega e_{i-1} = p_i + (\omega - 1)e_{i-1} = p_i + \omega^2 e_{i-1},$$

so $q_{i-1} \neq p_i$. Also $q_i = p_i + \omega e_i \neq p_i$. Further,

$$q_{i-1} - p_{i-1} = \omega e_{i-1} \neq 0,$$

and

$$q_i - p_{i+1} = (\omega - 1)e_i = \omega^2 e_i \neq 0.$$

If $q_i = p_{i-1}$, then

$$p_i + \omega e_i = p_i - e_{i-1},$$

so $e_i = -\omega^{-1}e_{i-1} = e^{2\pi i/3}e_{i-1}$, a counterclockwise turn $2\pi/3$, impossible. If $q_{i-1} = p_{i+1}$, then $e_i = \omega^2 e_{i-1}$, again a counterclockwise turn $2\pi/3$. If $q_{i-1} = q_i$, then $e_i = \omega e_{i-1}$, a counterclockwise turn $\pi/3$, impossible. Thus the four displayed neighbours are distinct. Since $\deg(p_i) = 4$, they are exactly all neighbours of p_i .

For p_0 , Swanepoel gives $\deg(p_0) = 3$ and $p_1, q_0 \in N(p_0)$. Let p_{-1} denote the third neighbour. For p_{11} , Swanepoel gives $\deg(p_{11}) = 4$ and $p_{10}, q_{10} \in N(p_{11})$, leaving at most two further neighbours. $\square$

Lemma 12 (Endpoint q -caps).

$$N(q_1) \subseteq \{p_1, p_2, q_0, q_2, t_1\} \cup U_1, \quad |U_1| \leq 1.$$

Also,

$$N(q_9) \subseteq \{p_9, p_{10}, q_8, q_{10}, t_8\} \cup U_9, \quad |U_9| \leq 1.$$

Proof. Swanepoel Lemma 8 applies to q_i for $1 \leq i \leq 2m - 5$. For $m = 7$, this includes q_1 and q_9 .

For q_1 , the two extra neighbours outside $\{p_1, p_2, q_0, q_2\}$ include $t_1 = s_1 = r_2$. Therefore at most one further extra neighbour remains, giving the stated U_1 .

For q_9 , the two extra neighbours outside $\{p_9, p_{10}, q_8, q_{10}\}$ include $t_8 = s_8 = r_9$. Therefore at most one further extra neighbour remains, giving the stated U_9 . $\square$

8. Separation lemmas

Lemma 13 (Boundary-boundary separation). *If $b - a \geq 2$, then*

$$|p_b - p_a| > 1.$$

Proof. If $b = a + 2$, then

$$p_b - p_a = e_a + e_{a+1},$$

so

$$|p_b - p_a| = 2 \cos(\tau_{a+1}/2) > \sqrt{3} > 1.$$

If $b - a \geq 3$, then

$$p_b - p_a = e_a + e_{a+1} + \cdots + e_{b-1}.$$

Project onto e_a . The first three terms contribute 1, $> 1/2$, and $> 1/2$, respectively. The projection is $> 2 > 1$, so the length is > 1 . $\square$

Lemma 14 (Boundary-to- q separation). *If $a \notin \{b, b+1\}$, then*

$$|p_a - q_b| > 1.$$

Proof. If $a < b$, then

$$q_b - p_a = e_a + \cdots + e_{b-1} + \omega e_b.$$

Project onto e_a . The e_a -term contributes 1, and the ωe_b -term contributes at least $1/2$. Thus the projection is > 1 .

If $a > b+1$, then

$$p_a - q_b = e_b + \cdots + e_{a-1} - \omega e_b = \bar{\omega} e_b + e_{b+1} + \cdots + e_{a-1}.$$

For $a = b+2$, the vector is $\bar{\omega} e_b + e_{b+1}$, whose two summands have angle $\pi/3 - \tau_{b+1} \in [0, \pi/3]$. Hence the squared length is at least 3. For $a \geq b+3$, project onto e_b : the first three terms contribute $1/2$, $> 1/2$, and $> 1/2$, so the projection is $> 3/2 > 1$. $\square$

Lemma 15 (Boundary-to- t separation). *Let*

$$t_b = p_b + \omega(e_b + e_{b+1}).$$

For every boundary vertex p_a ,

$$|p_a - t_b| > 1.$$

Proof. If $a < b$, then

$$t_b - p_a = e_a + \cdots + e_{b-1} + \omega e_b + \omega e_{b+1}.$$

Project onto e_a . The first e -term contributes 1, and the two ωe -terms each contribute at least $1/2$. Hence the projection is at least $2 > 1$.

If $a = b$, then

$$t_b - p_b = \omega(e_b + e_{b+1}),$$

whose length is $|e_b + e_{b+1}| > \sqrt{3} > 1$.

If $a = b+1$, then

$$p_{b+1} - t_b = \bar{\omega} e_b - \omega e_{b+1}.$$

Writing $e_{b+1} = e_b e^{-i\tau_{b+1}}$, its squared length is

$$2 - 2 \cos\left(\frac{2\pi}{3} - \tau_{b+1}\right) > 1,$$

since $0 \leq \tau_{b+1} < \pi/3$.

If $a = b+2$, then

$$p_{b+2} - t_b = \bar{\omega}(e_b + e_{b+1}),$$

whose length is $> \sqrt{3}$.

If $a > b+2$, then

$$p_a - t_b = \bar{\omega} e_b + \bar{\omega} e_{b+1} + e_{b+2} + \cdots + e_{a-1}.$$

Project onto e_{b+2} . The terms e_{b+2} , $\bar{\omega} e_{b+1}$, and $\bar{\omega} e_b$ contribute at least 1, $1/2$, and $1/2$, respectively; all later terms have nonnegative projection. Hence the projection is at least $2 > 1$. $\square$

Lemma 16 ($q_7 t_5$ separation). *If $t_4 t_5 \notin E(G)$, then*

$$|q_7 - t_5| > 1.$$

Proof. We prove $|q_7 - t_5| \geq \sqrt{3}$. We have

$$q_7 = p_7 + \omega e_7, \quad t_5 = p_5 + \omega(e_5 + e_6).$$

Therefore

$$q_7 - t_5 = e_5 + e_6 + \omega e_7 - \omega e_5 - \omega e_6 = \bar{\omega}(e_5 + e_6) + \omega e_7.$$

Rotate so $e_5 = 1$. Write

$$e_6 = e^{-ia}, \quad e_7 = e^{-i(a+b)},$$

with $a, b \geq 0$ and $a + b < \pi/3$. Multiplication by ω preserves length, so

$$|q_7 - t_5| = |1 + e^{-ia} + \omega^2 e^{-i(a+b)}|.$$

Let

$$F(a, b) = 1 + e^{-ia} + \omega^2 e^{-i(a+b)}.$$

Then

$$|F(a, b)|^2 = 3 + 2H(a, b),$$

where

$$H(a, b) = \cos a + \cos\left(\frac{2\pi}{3} - b\right) + \cos\left(\frac{2\pi}{3} - a - b\right).$$

For fixed a ,

$$\frac{\partial H}{\partial b} = \sin\left(\frac{2\pi}{3} - b\right) + \sin\left(\frac{2\pi}{3} - a - b\right) \geq 0,$$

because both arguments lie in $(\pi/3, 2\pi/3]$. Hence $H(a, b)$ is minimized at $b = 0$. Now

$$H(a, 0) = \cos a - \frac{1}{2} + \cos\left(\frac{2\pi}{3} - a\right) = \frac{1}{2} \cos a - \frac{1}{2} + \frac{\sqrt{3}}{2} \sin a.$$

Thus

$$2H(a, 0) = \cos a + \sqrt{3} \sin a - 1 = 2 \cos(a - \pi/3) - 1 \geq 0,$$

since $0 \leq a < \pi/3$. Therefore $H(a, b) \geq 0$, and $|F(a, b)|^2 \geq 3$. $\square$

9. Cap lemmas and flat-port geometry

Lemma 17 (Swanepoel t -caps).

$$|N(t_5) \setminus \{t_4, t_6, q_5, q_6\}| \leq 2,$$

$$|N(t_6) \setminus \{t_5, t_7, q_6, q_7\}| \leq 2,$$

$$|N(t_7) \setminus \{t_6, t_8, q_7, q_8\}| \leq 2.$$

Proof. Apply the Swanepoel Lemma 9 cap from Proposition 1 to the equality triples $(4, 5, 6)$, $(5, 6, 7)$, and $(6, 7, 8)$. For $t_5 = s_5$, the cap set is $\{s_4, s_6, q_5, q_6\} = \{t_4, t_6, q_5, q_6\}$. The other two cases are identical. $\square$

Definition 3 (Port and saturation). *In the fully flat branch, the t_i -port, for $i = 5, 6, 7$, is the closed angular interval of possible extra neighbours of t_i between the endpoint directions 60° and 120° relative to the flat strip direction. It is saturated if both endpoint positions*

$$x_i = t_i + \omega, \quad y_i = t_i + \omega^2$$

are occupied by vertices of the point set.

Lemma 18 (Fully flat port cap). *In the fully flat branch, every extra neighbour of t_i , $i = 5, 6, 7$, outside the strip-neighbour set lies in the t_i -port. If the port is saturated, its only port-neighbours are the two endpoints. If the port is not saturated, it contributes at most one extra neighbour. If one endpoint is already present and the port is not saturated, it contributes no further extra neighbour. Consecutive saturated ports share one endpoint.*

Proof. Place $t_i = (0, 0)$ with the flat strip direction horizontal. The four forced strip-neighbour directions at t_i are

$$0^\circ, \quad 180^\circ, \quad 240^\circ, \quad 300^\circ.$$

A further unit neighbour must have angular separation at least 60° from each. These four neighbours forbid the open arcs

$$\begin{aligned} &(-60^\circ, 60^\circ), \\ &(120^\circ, 240^\circ), \\ &(180^\circ, 300^\circ), \end{aligned}$$

and

$$(240^\circ, 360^\circ).$$

The only remaining arc is $[60^\circ, 120^\circ]$.

If two extra neighbours lie in this closed 60° interval, their angular separation is at most 60° . Since their mutual distance must be at least 1, the angular separation is exactly 60° ; hence they are the two endpoints. Thus a non-saturated port contributes at most one extra neighbour.

If one endpoint is present and another extra neighbour existed, it would either be an interior point, which is too close to the endpoint, or the other endpoint, which would saturate the port. Hence no further extra neighbour exists in the non-saturated case.

The sharing relation follows from

$$t_i + \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = t_{i+1} + \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

□

Lemma 19 (Branch-qualified degree caps). *In the fully flat branch, if the t_7 -port is saturated, then*

$$N(t_8) \subseteq \{q_8, q_9, t_7, x_7\} \cup U_8, \quad |U_8| \leq 2.$$

In the all-saturated branch,

$$N(x_6) \subseteq \{t_6, t_7, x_5, x_7\} \cup V_6, \quad |V_6| \leq 2.$$

If t_6, t_7 are saturated but t_5 is not saturated, then

$$N(x_6) \subseteq \{t_6, t_7, y_6, x_7\} \cup V_6, \quad |V_6| \leq 2.$$

Proof. In fully flat coordinates,

$$t_8 = 3 + 2\omega.$$

If t_7 is saturated, then

$$x_7 = 2 + 3\omega.$$

The vertices

$$q_8 = 3 + \omega, \quad q_9 = 4 + \omega, \quad t_7 = 2 + 2\omega, \quad x_7 = 2 + 3\omega$$

are distinct and have differences from t_8 equal to

$$-\omega, \quad 1 - \omega, \quad -1, \quad -1 + \omega,$$

all of length 1. Since $\deg(t_8) \leq 6$, at most two further neighbours exist.

For $x_6 = 1 + 3\omega$ in the all-saturated branch, the vertices

$$t_6 = 1 + 2\omega, \quad t_7 = 2 + 2\omega, \quad x_5 = 3\omega, \quad x_7 = 2 + 3\omega$$

are distinct and have differences from x_6 equal to

$$-\omega, \quad 1 - \omega, \quad -1, \quad 1.$$

All have length 1, so at most two further neighbours exist.

If t_6, t_7 are saturated but t_5 is not, then the left endpoint adjacent to x_6 is the actual vertex $y_6 = 3\omega$, not an actual saturated-port vertex x_5 . The four known neighbours are

$$t_6, t_7, y_6, x_7,$$

again with differences $-\omega, 1 - \omega, -1$, and 1 from x_6 . The degree-six bound gives the cap. $\square$

Lemma 20 (Neighbourhood inclusion). *Let S be independent. Suppose*

$$N(v) \subseteq A \cup C_1 \cup \dots \cup C_r \cup S$$

for every $v \in S$. Then

$$N(S) \subseteq A \cup C_1 \cup \dots \cup C_r.$$

Thus

$$|N(S)| \leq |A| + \sum_{j=1}^r |C_j|.$$

Proof. If $x \in N(S)$, then $x \notin S$ and $x \in N(v)$ for some $v \in S$. By the assumed containment, $x \in A \cup C_1 \cup \dots \cup C_r \cup S$. Since $x \notin S$, $x \in A \cup C_1 \cup \dots \cup C_r$. $\square$

10. The $t_4 t_5$ -absent branch

Lemma 21. *If $t_4 t_5 \notin E(G)$, then*

$$S_0 = \{p_0, p_2, p_4, p_6, p_9, q_7, t_5\}$$

is independent and

$$|N(S_0)| \leq 20.$$

Proof. The selected boundary vertices are mutually nonadjacent by Lemma 13. They are nonadjacent to q_7 by Lemma 14 and to t_5 by Lemma 15. Finally, $q_7 t_5$ is absent by Lemma 16. Thus S_0 is independent.

The selected boundary vertices contribute only

$$p_{-1}, p_1, p_3, p_5, p_7, p_8, p_{10}, q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_8, q_9.$$

The vertex q_7 contributes only

$$p_7, p_8, q_6, q_8, t_6, t_7.$$

The vertex t_5 has displayed possible neighbours t_4, t_6, q_5, q_6 . Since t_4t_5 is absent, the displayed neighbours reduce to t_6, q_5, q_6 , plus a Swanepoel cap C_5 with $|C_5| \leq 2$.

Let

$$A_0 = \{p_{-1}, p_1, p_3, p_5, p_7, p_8, p_{10}, q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_8, q_9, t_6, t_7\}.$$

Then $|A_0| \leq 18$ and

$$N(S_0) \subseteq A_0 \cup C_5.$$

Therefore

$$|N(S_0)| \leq 18 + 2 = 20.$$

□

11. The t_4t_5 -present branch before full flatness

Assume throughout this section that $t_4t_5 \in E(G)$. By Corollary 1,

$$\tau_5 = \tau_6 = 0.$$

Lemma 22. *If $t_4t_5 \in E(G)$ and $\tau_7 > 0$, then*

$$S_1 = \{p_6, p_8, q_4, t_5, t_6, t_7\}$$

is independent and

$$|N(S_1)| \leq 17.$$

Proof. The boundary pair p_6, p_8 is nonadjacent by Lemma 13. The vertices p_6, p_8 are nonadjacent to q_4 by Lemma 14 and to t_5, t_6, t_7 by Lemma 15.

The unique break inequality gives

$$\tau_3 + 2\tau_4 + \tau_5 \geq \frac{\pi}{3}.$$

Since $\tau_5 = 0$ and the total turn is $< \pi/3$, we get $\tau_4 > 0$. By the general q -turn criterion, Lemma 7,

$$q_3q_4 \notin E(G).$$

Swanepoel Lemma 8 gives

$$N(q_4) \subseteq \{p_4, p_5, q_3, q_5, r_4, s_4\}.$$

Since q_3q_4 is absent and $s_4 = t_4$,

$$N(q_4) \subseteq \{p_4, p_5, q_5, r_4, t_4\}.$$

Normalize $p_4 = 0$ and $e_4 = e_5 = e_6 = 1$. Let $a = \tau_7 > 0$ and $b = \tau_8 \geq 0$. Then $a + b < \pi/3$ and

$$e_7 = e^{-ia}, \quad e_8 = e^{-i(a+b)}.$$

Thus

$$q_4 = \omega, \quad t_5 = 1 + 2\omega, \quad t_6 = 2 + \omega(1 + e^{-ia}), \quad t_7 = 3 + \omega(e^{-ia} + e^{-i(a+b)}).$$

Now

$$t_5 - q_4 = 1 + \omega, \quad |t_5 - q_4| = \sqrt{3} > 1.$$

Also,

$$t_6 - q_4 = 2 + \omega e^{-ia},$$

whose real part is $2 + \cos(\pi/3 - a) > 1$. Similarly,

$$t_7 - q_4 = 3 - \omega + \omega e^{-ia} + \omega e^{-i(a+b)},$$

whose real part is

$$3 - \frac{1}{2} + \cos(\pi/3 - a) + \cos(\pi/3 - a - b) > 1.$$

By Lemma 8, $t_5 t_6$ is absent because $\tau_6 + \tau_7 > 0$, and $t_6 t_7$ is absent because $\tau_7 + \tau_8 > 0$. Finally,

$$t_7 - t_5 = 2\bar{\omega} + \omega(e^{-ia} + e^{-i(a+b)}).$$

Projection in the e_5 direction gives

$$2 \cos(\pi/3) + \cos(\pi/3 - a) + \cos(\pi/3 - a - b) > 2,$$

so $|t_7 - t_5| > 1$. Thus S_1 is independent.

Let

$$A_1 = \{p_4, p_5, p_7, p_9, q_5, q_6, q_7, q_8, r_4, t_4, t_8\}.$$

Then $|A_1| \leq 11$. The non-capped neighbours of the selected vertices lie in A_1 . The vertices t_5, t_6, t_7 have Swanepoel caps C_5, C_6, C_7 , each of size at most 2. Therefore

$$N(S_1) \subseteq A_1 \cup C_5 \cup C_6 \cup C_7,$$

and

$$|N(S_1)| \leq 11 + 2 + 2 + 2 = 17.$$

□

Lemma 23. *If $t_4 t_5 \in E(G)$, $\tau_7 = 0$, and $\tau_8 + \tau_9 > 0$, then*

$$S_2 = \{p_0, p_3, p_5, p_8, q_1, q_6, t_7\}$$

is independent and

$$|N(S_2)| \leq 20.$$

Proof. The selected boundary vertices are mutually nonadjacent by Lemma 13. They are nonadjacent to q_1, q_6 by Lemma 14 and to t_7 by Lemma 15.

Rotate so $e_1 = 1$. Then

$$q_6 - q_1 = e_1 + e_2 + e_3 + e_4 + e_5 + \omega e_6 - \omega e_1.$$

By Lemma 9,

$$\Re(q_6 - q_1) \geq 5 \cdot \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{5}{2} > 1.$$

Thus $q_1 q_6$ is absent.

Next,

$$t_7 - q_1 = e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + \omega e_7 + \omega e_8 - \omega e_1.$$

Thus

$$\Re(t_7 - q_1) \geq 6 \cdot \frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{7}{2} > 1.$$

So $q_1 t_7$ is absent.

Since $\tau_6 = \tau_7 = 0$, we have $e_6 = e_7$, and

$$t_7 - q_6 = e_6 + \omega e_8.$$

The angle between e_6 and ωe_8 is $\pi/3 - \tau_8$. Hence

$$|t_7 - q_6|^2 = 2 + 2 \cos(\pi/3 - \tau_8) \geq 3 > 1.$$

Thus $q_6 t_7$ is absent, and S_2 is independent.

By Lemma 12,

$$N(q_1) \subseteq \{p_1, p_2, q_0, q_2, t_1\} \cup U_1, \quad |U_1| \leq 1.$$

Also,

$$N(q_6) \subseteq \{p_6, p_7, q_5, q_7, t_5, t_6\}.$$

Since $\tau_8 + \tau_9 > 0$, Lemma 8 gives $t_7 t_8 \notin E(G)$. The Swanepoel cap gives

$$N(t_7) \subseteq \{q_7, q_8, t_6\} \cup C_7, \quad |C_7| \leq 2.$$

Let

$$A_2 = \{p_{-1}, p_1, p_2, p_4, p_6, p_7, p_9, q_0, q_2, q_3, q_4, q_5, q_7, q_8, t_1, t_5, t_6\}.$$

Then $|A_2| \leq 17$ and

$$N(S_2) \subseteq A_2 \cup U_1 \cup C_7.$$

Therefore

$$|N(S_2)| \leq 17 + 1 + 2 = 20.$$

□

12. Fully flat branch

Assume

$$\tau_7 = \tau_8 = \tau_9 = 0.$$

Normalize

$$p_5 = 0, \quad e_5 = e_6 = e_7 = e_8 = e_9 = 1.$$

Then

$$\begin{aligned} p_6 &= 1, & p_7 &= 2, & p_8 &= 3, & p_9 &= 4, & p_{10} &= 5, \\ q_5 &= \omega, & q_6 &= 1 + \omega, & q_7 &= 2 + \omega, & q_8 &= 3 + \omega, & q_9 &= 4 + \omega, \end{aligned}$$

and

$$t_5 = 2\omega, \quad t_6 = 1 + 2\omega, \quad t_7 = 2 + 2\omega, \quad t_8 = 3 + 2\omega.$$

Also

$$p_{11} = 5 + e^{-i\theta}, \quad 0 \leq \theta < \frac{\pi}{3}.$$

When a port is saturated,

$$x_i = t_i + \omega, \quad y_i = t_i + \omega^2.$$

Thus

$$x_5 = 3\omega, \quad x_6 = 1 + 3\omega, \quad x_7 = 2 + 3\omega,$$

and

$$y_6 = x_5, \quad y_7 = x_6.$$

Lemma 24. *If t_5 and t_7 are saturated, then t_6 is saturated. Hence the possible non-all-saturated patterns are exactly*

$$\emptyset, \quad t_5, \quad t_6, \quad t_7, \quad t_5t_6, \quad t_6t_7.$$

Proof. Saturation at t_5 supplies the endpoint $x_5 = y_6$ of the t_6 -port. Saturation at t_7 supplies the endpoint $x_6 = y_7$ of the t_6 -port. Thus both endpoints of the t_6 -port are present, so t_6 is saturated. $\square$

Lemma 25 (No saturated port). *If none of the ports at t_5, t_6, t_7 is saturated, then the set*

$$S_\emptyset = \{p_6, p_8, p_{11}, q_9, t_5, t_7\}$$

is independent and satisfies $|N(S_\emptyset)| \leq 17$.

Proof. The non- p_{11} selected-pair differences are

$$p_8 - p_6 = 2,$$

$$q_9 - p_6 = 3 + \omega,$$

$$t_5 - p_6 = 2\omega - 1,$$

$$t_7 - p_6 = 1 + 2\omega,$$

$$q_9 - p_8 = 1 + \omega,$$

$$t_5 - p_8 = 2\omega - 3,$$

$$t_7 - p_8 = 2\omega - 1,$$

$$t_5 - q_9 = \omega - 4,$$

$$t_7 - q_9 = \omega - 2,$$

and

$$t_7 - t_5 = 2.$$

None is 0 or one of the six triangular-lattice unit vectors. Since the graph has minimum distance 1, these differences all have length > 1 , except where the displayed length is visibly 2 or $\sqrt{3}$.

For p_{11} ,

$$p_{11} - q_9 = 1 + e^{-i\theta} - \omega,$$

with real part $1 + \cos \theta - 1/2 > 1$. Also,

$$p_{11} - t_5 = 5 + e^{-i\theta} - 2\omega,$$

with real part $4 + \cos \theta > 1$,

$$p_{11} - t_7 = 3 + e^{-i\theta} - 2\omega,$$

with real part $2 + \cos \theta > 1$,

$$p_{11} - p_6 = 4 + e^{-i\theta},$$

with real part $4 + \cos \theta > 1$, and

$$p_{11} - p_8 = 2 + e^{-i\theta},$$

with real part $2 + \cos \theta > 1$. Thus $S_\emptyset$ is independent.

Let

$$A_\emptyset = \{p_5, p_7, p_9, p_{10}, q_5, q_6, q_7, q_8, q_{10}, t_4, t_6, t_8\}.$$

Then $|A_\emptyset| \leq 12$ and

$$N(S_\emptyset) \subseteq A_\emptyset \cup U_{11} \cup U_9 \cup C_5 \cup C_7,$$

where $|U_{11}| \leq 2$, $|U_9| \leq 1$, and $|C_5|, |C_7| \leq 1$ are the flat-port caps for the unsaturated t_5 and t_7 ports. Hence

$$|N(S_\emptyset)| \leq 12 + 2 + 1 + 1 + 1 = 17.$$

□

Lemma 26 (Only t_5 saturated). *If only the t_5 -port is saturated, then*

$$S_5 = \{p_5, p_8, p_{11}, q_6, q_9, t_7\}$$

is independent and satisfies $|N(S_5)| \leq 17$.

Proof. The selected-pair differences not involving p_{11} are

$$\begin{aligned} p_5 - p_8 &= -3, \\ p_5 - q_6 &= -(1 + \omega), \\ p_5 - q_9 &= -(4 + \omega), \\ p_5 - t_7 &= -(2 + 2\omega), \\ p_8 - q_6 &= 2 - \omega, \\ p_8 - q_9 &= -(1 + \omega), \\ p_8 - t_7 &= 1 - 2\omega, \\ q_6 - q_9 &= -3, \\ q_6 - t_7 &= -(1 + \omega), \end{aligned}$$

and

$$q_9 - t_7 = 2 - \omega.$$

None is 0 or a triangular-lattice unit vector, and the cases of lengths 3 or $\sqrt{3}$ are immediate.

For p_{11} ,

$$p_{11} - p_5 = 5 + e^{-i\theta},$$

whose real part is $5 + \cos \theta > 1$;

$$p_{11} - p_8 = 2 + e^{-i\theta},$$

whose real part is $2 + \cos \theta > 1$;

$$p_{11} - q_6 = 4 + e^{-i\theta} - \omega,$$

whose real part is $4 + \cos \theta - 1/2 > 1$;

$$p_{11} - q_9 = 1 + e^{-i\theta} - \omega,$$

whose real part is $1 + \cos \theta - 1/2 > 1$; and

$$p_{11} - t_7 = 3 + e^{-i\theta} - 2\omega,$$

whose real part is $2 + \cos \theta > 1$. Thus S_5 is independent.

Let

$$A_5 = \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8\}.$$

Then $|A_5| \leq 13$ and

$$N(S_5) \subseteq A_5 \cup U_{11} \cup U_9 \cup C_7,$$

where $|U_{11}| \leq 2$, $|U_9| \leq 1$, and $|C_7| \leq 1$. Hence

$$|N(S_5)| \leq 13 + 2 + 1 + 1 = 17.$$

□

Lemma 27 (Only t_6 saturated). *If only the t_6 -port is saturated, then*

$$S_6 = \{p_5, p_8, p_{11}, q_6, q_9, t_7\}$$

is independent and satisfies $|N(S_6)| \leq 17$.

Proof. This is the same selected set as in Lemma 26, so it is independent. Let

$$A_6 = \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8, x_6\}.$$

Then $|A_6| \leq 14$. The t_7 -port is not saturated, but the saturated t_6 -port supplies one endpoint $x_6 = y_7$ of the t_7 -port. By Lemma 18, t_7 has no further port neighbour. Therefore

$$N(S_6) \subseteq A_6 \cup U_{11} \cup U_9,$$

and

$$|N(S_6)| \leq 14 + 2 + 1 = 17.$$

□

Lemma 28 (Only t_7 saturated). *If only the t_7 -port is saturated, then*

$$S_7 = \{p_0, p_3, p_6, q_1, q_4, t_5\}$$

is independent and satisfies $|N(S_7)| \leq 17$.

Proof. Boundary-boundary nonadjacency follows from Lemma 13; boundary-to- q and boundary-to- t nonadjacency follow from Lemmas 14 and 15.

The remaining auxiliary pairs are q_1q_4, q_1t_5, q_4t_5 . First,

$$q_4 - q_1 = e_1 + e_2 + e_3 + \omega e_4 - \omega e_1,$$

so

$$\Re(q_4 - q_1) \geq \frac{3}{2} > 1.$$

Thus q_1q_4 is absent. Next,

$$t_5 - q_1 = e_1 + e_2 + e_3 + e_4 + \omega e_5 + \omega e_6 - \omega e_1,$$

so

$$\Re(t_5 - q_1) \geq \frac{5}{2} > 1.$$

Thus $q_1 t_5$ is absent. Finally, since $e_4 = e_5 = e_6$, normalizing $p_4 = 0$ gives

$$q_4 = \omega, \quad t_5 = 1 + 2\omega.$$

Thus

$$t_5 - q_4 = 1 + \omega,$$

and the distance is $\sqrt{3} > 1$. Hence S_7 is independent.

Let

$$A_7 = \{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_4, t_1, t_4, t_6\}.$$

Then $|A_7| \leq 15$ and

$$N(S_7) \subseteq A_7 \cup U_1 \cup C_5,$$

with $|U_1| \leq 1$ and $|C_5| \leq 1$. Therefore

$$|N(S_7)| \leq 15 + 1 + 1 = 17.$$

□

Lemma 29 (t_5, t_6 saturated). *If the t_5 - and t_6 -ports are saturated and the t_7 -port is not saturated, then*

$$S_{56} = \{p_5, p_8, p_{11}, q_6, q_9, t_7\}$$

is independent and satisfies $|N(S_{56})| \leq 17$.

Proof. The selected set is the same as in Lemma 26, so it is independent. Let

$$A_{56} = \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8, x_6\}.$$

Then $|A_{56}| \leq 14$. The t_7 -port is not saturated, and the saturated t_6 -port supplies the endpoint $x_6 = y_7$. By Lemma 18, t_7 has no further port neighbour. Hence

$$N(S_{56}) \subseteq A_{56} \cup U_{11} \cup U_9,$$

and

$$|N(S_{56})| \leq 14 + 2 + 1 = 17.$$

□

Lemma 30 (t_6, t_7 saturated). *If the t_6 - and t_7 -ports are saturated and the t_5 -port is not saturated, then*

$$S_{67} = \{p_6, p_9, q_7, t_5, t_8, x_6\}$$

is independent and satisfies $|N(S_{67})| \leq 17$.

Proof. In fully flat coordinates,

$$p_6 = 1, \quad p_9 = 4, \quad q_7 = 2 + \omega, \quad t_5 = 2\omega, \quad t_8 = 3 + 2\omega, \quad x_6 = 1 + 3\omega.$$

The selected-pair differences are

$$p_9 - p_6 = 3,$$

$$q_7 - p_6 = 1 + \omega,$$

$$t_5 - p_6 = 2\omega - 1,$$

$$\begin{aligned}
t_8 - p_6 &= 2 + 2\omega, \\
x_6 - p_6 &= 3\omega, \\
q_7 - p_9 &= -2 + \omega, \\
t_5 - p_9 &= 2\omega - 4, \\
t_8 - p_9 &= -1 + 2\omega, \\
x_6 - p_9 &= -3 + 3\omega, \\
t_5 - q_7 &= \omega - 2, \\
t_8 - q_7 &= 1 + \omega, \\
x_6 - q_7 &= -1 + 2\omega, \\
t_8 - t_5 &= 3, \\
x_6 - t_5 &= 1 + \omega, \\
x_6 - t_8 &= -2 + \omega.
\end{aligned}$$

None is 0 or one of $\pm 1, \pm\omega, \pm\omega^2$. Thus no selected distance is 1, and all selected distances are > 1 . Hence S_{67} is independent.

Let

$$A_{67} = \{p_5, p_7, p_8, p_{10}, q_5, q_6, q_8, q_9, t_4, t_6, t_7, y_6, x_7\}.$$

Then $|A_{67}| \leq 13$. Since t_7 is saturated, Lemma 19 gives

$$N(t_8) \subseteq \{q_8, q_9, t_7, x_7\} \cup U_8, \quad |U_8| \leq 2.$$

Since t_6, t_7 are saturated but t_5 is not, Lemma 19 gives

$$N(x_6) \subseteq \{t_6, t_7, y_6, x_7\} \cup V_6, \quad |V_6| \leq 2.$$

Finally, $y_6 = x_5$ occupies one endpoint of the t_5 -port, and t_5 is not saturated, so Lemma 18 gives no further t_5 -port neighbour. Therefore

$$N(S_{67}) \subseteq A_{67} \cup U_8 \cup V_6,$$

and

$$|N(S_{67})| \leq 13 + 2 + 2 = 17.$$

□

Corollary 2. *In the fully flat branch, if the ports at t_5, t_6, t_7 are not all saturated, then there is an independent set S with*

$$|S| = 6, \quad |N(S)| \leq 17.$$

Proof. By Lemma 24, the non-all-saturated patterns are exactly the six patterns in Lemmas 25–30. Each lemma supplies the required set. □

13. All-saturated branch

Assume the ports at t_5, t_6, t_7 are all saturated. Let

$$S_3 = \{p_0, p_3, p_6, p_9, q_1, q_4, q_7, t_5, t_8, x_6\}.$$

Lemma 31. *The set S_3 is independent and satisfies*

$$|N(S_3)| \leq 28.$$

Proof. Boundary-boundary nonadjacency follows from Lemma 13; boundary-to- q and boundary-to- t nonadjacency follow from Lemmas 14 and 15.

We check x_6 against selected boundary vertices. In the fully flat branch,

$$x_6 = p_6 + 3\omega e_6.$$

Thus

$$|x_6 - p_6| = 3.$$

In flat coordinates,

$$x_6 - p_9 = -3 + 3\omega,$$

so $|x_6 - p_9| = 3$. For p_3 ,

$$x_6 - p_3 = e_3 + e_4 + e_5 + 3\omega e_6.$$

Rotating so $e_3 = 1$ gives

$$\Re(x_6 - p_3) > 1 + \frac{1}{2} + \frac{1}{2} + 3 \cdot \frac{1}{2} = \frac{7}{2} > 1.$$

For p_0 ,

$$x_6 - p_0 = e_0 + e_1 + e_2 + e_3 + e_4 + e_5 + 3\omega e_6.$$

Rotating so $e_0 = 1$ gives

$$\Re(x_6 - p_0) > 1 + 5 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = 5 > 1.$$

Thus x_6 is not adjacent to the selected boundary vertices.

Now consider pairs involving q_1 :

$$q_4 - q_1 = e_1 + e_2 + e_3 + \omega e_4 - \omega e_1,$$

so $\Re(q_4 - q_1) \geq 3/2 > 1$. Also,

$$q_7 - q_1 = e_1 + \cdots + e_6 + \omega e_7 - \omega e_1,$$

so $\Re(q_7 - q_1) \geq 3 > 1$. Further,

$$t_5 - q_1 = e_1 + \cdots + e_4 + \omega e_5 + \omega e_6 - \omega e_1,$$

so $\Re(t_5 - q_1) \geq 5/2 > 1$. Also,

$$t_8 - q_1 = e_1 + \cdots + e_7 + \omega e_8 + \omega e_9 - \omega e_1,$$

so $\Re(t_8 - q_1) \geq 7/2 > 1$. Finally,

$$x_6 - q_1 = e_1 + \cdots + e_5 + 3\omega e_6 - \omega e_1,$$

so $\Re(x_6 - q_1) \geq 7/2 > 1$.

Now check pairs involving q_4 . The break inequality gives

$$\tau_3 + 2\tau_4 + \tau_5 \geq \frac{\pi}{3}.$$

Since $\tau_5 = 0$ and total turn is $< \pi/3$, we get $\tau_4 > 0$. By the general q -turn criterion, $q_3q_4 \notin E(G)$. In the fully flat normalization,

$$p_4 = -1, \quad q_4 = -1 + \omega.$$

Also,

$$q_7 = 2 + \omega, \quad t_5 = 2\omega, \quad t_8 = 3 + 2\omega, \quad x_6 = 1 + 3\omega.$$

Thus

$$\begin{aligned} q_7 - q_4 &= 3, \\ t_5 - q_4 &= 1 + \omega, \\ t_8 - q_4 &= 4 + \omega, \\ x_6 - q_4 &= 2 + 2\omega. \end{aligned}$$

All have length greater than 1.

Among q_7, t_5, t_8, x_6 , the differences are

$$\begin{aligned} t_5 - q_7 &= \omega - 2, \\ t_8 - q_7 &= 1 + \omega, \\ x_6 - q_7 &= -1 + 2\omega, \\ t_8 - t_5 &= 3, \\ x_6 - t_5 &= 1 + \omega, \\ x_6 - t_8 &= -2 + \omega. \end{aligned}$$

None is a triangular-lattice unit vector, so every selected-pair distance is > 1 . Hence S_3 is independent.

Now define

$$A_3 = \{p_{-1}, p_1, p_2, p_4, p_5, p_7, p_8, p_{10}, q_0, q_2, q_3, q_5, q_6, q_8, q_9, \\ r_4, t_1, t_4, t_6, t_7, x_5, x_7, y_5\}.$$

Then $|A_3| \leq 23$.

The local neighbourhood inclusions are

$$\begin{aligned} N(p_0), N(p_3), N(p_6), N(p_9) &\subseteq A_3, \\ N(q_1) &\subseteq A_3 \cup U_1, \quad |U_1| \leq 1. \end{aligned}$$

Since q_3q_4 is absent and $s_4 = t_4$,

$$N(q_4) \subseteq A_3.$$

Also,

$$N(q_7) \subseteq A_3, \quad N(t_5) \subseteq A_3.$$

Since all ports are saturated, Lemma 19 gives

$$N(t_8) \subseteq A_3 \cup U_8, \quad |U_8| \leq 2,$$

and

$$N(x_6) \subseteq A_3 \cup V_6, \quad |V_6| \leq 2.$$

Thus

$$N(S_3) \subseteq A_3 \cup U_1 \cup U_8 \cup V_6.$$

Consequently,

$$|N(S_3)| \leq 23 + 1 + 2 + 2 = 28.$$

□

14. Final proof

Proof of Theorem 1. Assume that G is a smallest counterexample. By Lemma 1, every nonempty independent set S in G must satisfy

$$|N(S)| > \left\lfloor \frac{20|S|}{7} \right\rfloor.$$

By Lemma 2, the unique equality break is $s_3 \neq r_4$, and t_i is defined for $i = 1, 2, 4, 5, 6, 7, 8$.

If $t_4 t_5 \notin E(G)$, Lemma 21 gives an independent set S_0 with $|S_0| = 7$ and

$$|N(S_0)| \leq 20 = \left\lfloor \frac{20 \cdot 7}{7} \right\rfloor,$$

contradiction. Hence $t_4 t_5 \in E(G)$.

By Corollary 1, $\tau_5 = \tau_6 = 0$. If $\tau_7 > 0$, Lemma 22 gives an independent set S_1 with $|S_1| = 6$ and

$$|N(S_1)| \leq 17 = \left\lfloor \frac{20 \cdot 6}{7} \right\rfloor,$$

contradiction.

If $\tau_7 = 0$ and $\tau_8 + \tau_9 > 0$, Lemma 23 gives an independent set S_2 with $|S_2| = 7$ and

$$|N(S_2)| \leq 20 = \left\lfloor \frac{20 \cdot 7}{7} \right\rfloor,$$

contradiction.

Therefore $\tau_7 = \tau_8 = \tau_9 = 0$. If the ports at t_5, t_6, t_7 are not all saturated, Corollary 2 gives an independent set S with $|S| = 6$ and

$$|N(S)| \leq 17 = \left\lfloor \frac{20 \cdot 6}{7} \right\rfloor,$$

contradiction.

Thus all three ports are saturated. Lemma 31 gives an independent set S_3 with $|S_3| = 10$ and

$$|N(S_3)| \leq 28 = \left\lfloor \frac{20 \cdot 10}{7} \right\rfloor,$$

contradiction.

Every branch contradicts the deletion criterion. Therefore no smallest counterexample exists. Hence

$$\alpha(G) \geq \frac{7}{27}|V(G)|.$$

Since $\alpha(G)$ is an integer,

$$\alpha(G) \geq \left\lceil \frac{7|V(G)|}{27} \right\rceil.$$

Taking the minimum over all n -vertex Euclidean minimum-distance graphs gives

$$g(n) \geq \left\lceil \frac{7n}{27} \right\rceil.$$

□

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